User Administration :

GROUP:
Group is a collection of users , The purpose of the group is for easier administration of the users. For every group we have group name and group id.
Group Name is for User Reference

Group ID (GID) is for System Reference

By default 0 to 999 GID's are reserved for system reference. We can start creating the groups from GID 1000 +.

We have two types of groups:

1) Primary Group
2) Secondary Group

Primary Group :

If we wanted to create user he has to be a part of at least one primary group. Min is one and max also one. The same user cannot be a part of more than one primary group.

Secondary Group :

Its not mandatory the user has to be a part of secondary group but the same user can be a part of more than one secondary group up to a max of 15.

NOTE Single user can belongs to max 1 primary and 15 secondary groups & total he can belongs up to 16 groups.

NOTE:

The complete groups information is stored in /etc/group file

Filelds Present in /etc/group:

#vi /etc/group

<groupname> : X : <user>

X --→ Password place holder in /etc/gshadow file

:wq

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Groups encrypted password information is stored in /etc/gshadow file

Creating Group with default GID

#groupadd -g <GN>
g51

#cat /etc/group |grep g51
O/P

G51:x:1010

Creating group with custom GID:

#groupadd -g <GID> <GN>
2001 g52

Modifying GID:

#groupmod -g <NGID> <GN>
3001 g52

Renaming Group:

#groupmod -n <NGname> <OldGname>
G53 g52

Setting the password for the group
#gpasswd <GN>
G53

Adding single user to the group:
#gpasswd -a <UN> <GN>

Steve g53
Adding multiple users to a group :
#gpasswd -M u1,u2,u3,u4 <GN>
{ users } G53

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Deleting user from group:

#gpasswd -d <UN> <GN>

U1 g53

Deleting Group:

#groupdel <GN>
G1

Note: Any changes we do with the groups, All the changes are updated to /etc/group

USERS:

As same as group, For every user we have Unique UID and username. UID is for user reference and username is for kernel reference.
-→By default 0 to 999 UIDs are reserved for system reference. (0 is the Super user ID assigned for ROOT user)

UID 1 to 200 [These user IDS are reserved for System defined users)

UID 201 to 999 (These user Ids are assigned to the application related users)

We can start using the user id's starting from 1000+ for creating new users.

NOTE :

-→The complete users information is stored in /etc/passwd file

-→The users encrypted passwd Information is stored in /etc/shadow file

Fields present in /etc/passwd File

#vi /etc/passwd

Username : x : UID : PGID : User description : Home dir : Login Shell

:wq

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Fields present in /etc/shadow file
#vi /etc/shadow

Username : Encrypted Password : password Aging

:wq

Creating user with default attributes:

#useradd <UN>
U71

Creating user with custom attributes:

User Name – dom
P Group ---→ g81
S Groups ----→g86,g87

UID ---→4001

Comment ----→labadmin
Home Dir ------→ /erp/u71

Shell --→/bin/sh

Pre Req:

#groupadd g81

#groupadd g86

#groupadd g87

#mkdir /erp

#useradd -u 4001 -g g81 -G g86,g87 -c labadmin -d

/erp/u71 -m -s /bin/sh dom

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Setting the password

#passwd <UN>
Dom

Modifying user attributes :

User Name – dom → karan
P Group ---→ g81 -- →g91
S Groups ----→g86,g87-→additionally add one more group g88

UID ---→4001 --- →5001

Comment ----→labadmin→nwadmin
Home Dir ------→ /erp/dom → /corona/karan

Shell --→/bin/sh --→ /bin/bash

Pre req

#mkdir –p /corona/karan

#groupadd g91

#groupadd g88

Mod Users Primary group:

#usermod -g <NPG> <UN>
G91 dom

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Mod users Secondary group:

```
#usermod    -G    <Newgroup>    <UN>
                  G86,g87,g88   dom
```

Mod users Description:

```
#usermod    -c    <ND>    <UN>
                  Neadmin dom
```

Mod users Home dir:

```
#usermod    -d    <NHD>          <UN>
                  /corona/karan  dom
```

Mod USERID:

```
#usermod    -u    NUID>    <UN>
                  5001     dom
```

Renaming User:

```
#usermod    -l    <NUN    <OUN>
                  Karan   dom
```

NOTE : all these changes are updated to /etc/passwd

Deleting User:

```
#userdel    <UN>
            U1
```

Deleting the user along with Home dir:

```
#userdel    -r    <UN>
                  U2
```

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